

2078

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Architecture**

Code No: **CSC165** Semester: **3rd Semester**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Differentiate between hardwired and microprogrammed control unit. Describe with example of microprogram sequencer used in microprogrammed control unit.
2. Explain the various types of addressing modes and compare them algorithm, advantages and disadvantages.
3. Explain the non-restoring division algorithm with flow chart, now hardware implementation diagram. Divide 10/3 using restoring division.

Section B

Attempt any Eight questions[8×5=40]

4. Explain the bus interconnection scheme with diagram.
5. What do you mean by instruction format? Explain with an example.
6. Explain the data transfer instruction with example.
7. Explain the symbolic microinstructions with example.
8. Explain an instruction pipeline with suitable examples.
9. Explain the BOooth Multiplication algorithm with example.
10. What are the advantages and disadvantages of direct mapping and associative mapping between cache and main memory?
11. Differentiate between Input-output processor with Direct memory access.
12. Draw a three-dimensional hypercube and explain with example.